PET Florbetapir SUVRs (eroded WM reference region) - Insight46 Phase 1

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Document details

Categories of variables	PET Florbetapir SUVRs (eroded WM reference region)
Summary of work undertaken	Source data transferred to Stata .dta file • See below for summary of SUVRs
Names of source variables	Subject, ID, Scan, Date, SUVR, Amyloid, Status, SUVR, PVC, PVC, Amyloid, Status, Imputed
Output variables	suvr_i46p1, •, amyloidstatus_i46p1, •, suvrpvc_i46p1, •, pvcamyloidstatus_i46p1, •, imp_amyloid_i46p1, •, amyloid_comments_i46p1
Papers using these variables	Blood and cerebrospinal fluid biomarkers in Alzheimer's disease – from clinical to preclinical cohorts. PhD thesis (Ashvini Keshavan). • 2. Plasma amyloid, tau and serum neurofilament light chain in Insight 46 – associations with cognition and brain imaging. AAIC 2019 poster (Keshavan et al.) • 3. The Insight 46 Cohort – Relationships with Cerebral PET Amyloid positivity Using Two Independent Methods for Plasma Amyloid Beta Measurement. AAIC 2019 Featured Research Session Oral presentation. (Schott et al.) • 4. Associations between blood pressure across adulthood and late-life brain structure and pathology in the neuroscience substudy of the 1946 British birth cohort (Insight 46): an epidemiological study. The Lancet Neurology. (Lane et al. 2019)

Methods:

Adapted from: Lane, C. A., Parker, T. D., Cash, D. M., Macpherson, K., Donnachie, E., Murray-Smith, H., ... & Brown, D. (2017). Study protocol: Insight 46–a neuroscience sub-study of the mrc national survey of health and development. BMC neurology, 17(1), 75.

Image Acquisition:

Scanning was performed on a Biograph mMR 3 T PET/MRI scanner (Siemens Healthcare, Erlangen). Amyloid load is assessed using the 18F amyloid PET ligand, florbetapir. After intravenous cannulation, 370 MBq florbetapir F18 (Amyvid) is injected. PET data are acquired continuously during and following injection to allow florbetapir uptake dynamics to be assessed. Final amyloid burden is assessed over a 10-min period, ~50 min after injection, with scope for the previous 10-min period to be used if longer scan periods are not tolerated.

PET data, acquired in list-mode, is reconstructed using a 3D ordered-subset expectation-maximisation algorithm with three iterations and 21 subsets, and smoothed with a 4 mm Gaussian kernel. Attenuation maps are computed from the T1-weighted and T2-weighted volumetric scans using a multi-atlas CT synthesis method [76], also known as pseudo-CT (pCT).

Image Processing:

T1 images were parcellated using GIF v3 and then parcellations were registered to the PET images. Global standardised uptake value ratios (SUVRs) were calculated from a composite cortical target region normalised to an eroded white matter reference region and with and without partial volume correction (PVC) using the Discrete Iterative Yang technique. Positive or negative amyloid status was determined using a Gaussian mixture model applied to SUVRs, with the 99th percentile of the lower Gaussian as the cutoff (0·6104).

PLEASE NOTE that this pipeline has been performed for the cross-sectional analysis of SUVR only. SUVR will be computed in a longitudinally consistent fashion on both timepoints, and this may result in small changes in SUVR (<0.5% in most cases and < 2% in nearly all cases) when comparing values generated in cross-sectional pipeline with those in the longitudinal pipeline.

Variables in this document

5 medium-confidence matches

Variable	Label	How it was found
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Other medical conditions [51011]		
blood	Have you had anaemia in the last 10 years - at age 43 years	prose scan
Sub-Studies [57] › Insight46 [699] › Brain PET [6034] › Amyloid data [60341] › Cross-sectional beta-amyloid status (phase 1) [603411]		
amyloidstatus_i46p1	Amyloid-beta status with eroded white matter (WM) reference region - Insight 46 Phase 1	prose scan
imp_amyloid_i46p1	Imputed amyloid-beta status with eroded white matter (WM) reference region - Insight 46 Phase 1	prose scan
suvr_i46p1	Global standardised uptake value ratios (SUVR) with eroded white matter (WM) reference region - Insight 46 Phase 1	prose scan
suvrpvc_i46p1	Global standardised uptake value ratios (SUVR) with eroded white matter (WM) reference region with partial volume correction (PVC) - Insight 46 Phase 1	prose scan